INTERNATIONAL QUANTITATIVE INVESTING CHALLENGE 2026

JUNEJA STREET

Strategy Declaration

ML Supertrend: Adaptive Reversal Strategy

Young Investors Cup 2026 gold medallists

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Investment hypothesis and complete strategy

We target temporary selling-pressure dislocations in volatile stocks, buying after a fresh low and an upward reversal. The hypothesis fails when a decline reflects lasting information or trading costs absorb the rebound. Each completed 30-minute candle updates price, volume, ATR (average true range) and RSI (relative strength index). In the background, the model tests hypothetical long and short trades over five candles. Completed probes train the machine-learning models through rolling statistics, weighted batches and memory of similar conditions. The models adjust band width and smoothing length in bounded steps. A fresh low, upward trend flip, RSI(14) below 30 within 50 candles and a ten-candle gap since the last signal produce a BUY. We process earlier signals first, with alphabetical ticker order for ties. Position size is the smaller of the 15% value cap and the 0.5% planned-loss cap, using a stop two ATR(24) below entry. Liquidity, sector and remaining portfolio limits can reduce the order further. We hold at most five long positions, with 15% per name and 75% maximum total stock exposure. Aggregate planned stop risk remains 2%. We enter after confirmation and close on SELL, a protective stop or an account loss trigger.

A probe records what a hypothetical trade would have earned from the candle open to the close of its fifth candle. A rise of one entry ATR gives the long probe a score of +1 and the short probe -1. The model uses the result only after the fifth candle closes. Probes overlap, so their results are dependent.

Batch weights equal confidence / (1 + adverse movement in ATR units): one ATR of adverse movement halves the weight at equal confidence. Context memory contributes up to 65% of the context/global blend. The model smooths the resulting target; the worked update appears below. Learning changes signal sensitivity; position sizes follow the separate loss budget.

Why these stocks

DELL is the paper-trading example. For risk calibration, we use a separate historical cross-sector sample. It is not the strategy portfolio or a list of declared holdings. The trading universe remains broader: each stock must pass its BUY, liquidity and risk checks before receiving capital.

1 Strategy and quantitative method

Use completed regular-session 30-minute candles for supported S&P 500 stocks. BUY opens one long position; repeated BUYs do not add size. SELL or an earlier protective stop closes it. Orders follow confirmation; historical replay fills at the next bar open plus costs.

1. Measure normal price movement

True Range Equation

$$TR_t = \max\{H_t - L_t,\ |H_t - C_{t-1}|,\ |L_t - C_{t-1}|\}$$

t is the current completed candle; t - 1 is the previous candle. H, L and C are high, low and close in dollars. TR takes the largest candle range or gap from the previous close. Vertical bars mean absolute value.

Volatility Envelope Equations

$$x_t = \frac{H_t + L_t + 2C_t}{4}, \qquad A_t = A_{t-1} + \frac{TR_t - A_{t-1}}{\ell_t}$$

$$b_t^- = x_t - m_t A_t, \qquad b_t^+ = x_t + m_t A_t$$

x is the close-weighted price, A the smoothed true range, l the smoothing length and m the band multiplier. The raw boundaries are b minus and b plus. Initially l = 24 and m = 1.4, so each raw boundary lies 1.4 times A from x. Doubling A doubles that distance. The lower boundary can only rise while the previous close is above it; the upper can only fall while that close is below it. Otherwise that boundary resets. Crossing the prior opposing boundary flips the trend.

2. Machine-learning models and completed probes

Probe Return Equation

$$g = s\frac{C - P}{A_{\text{entry}}}$$

g is the probe score; s is +1 for a long and -1 for a short. P is entry price, C the fifth-candle close and A entry the smoothed true range at entry. A one-ATR price rise gives g = +1 for the long and -1 for the short. Up to five probes per direction can overlap; each result enters learning after its fifth candle closes.

Three feedback routes propose settings. Rolling results propose wider, slower bands below a 38% win rate with a negative average score, and tighter, faster bands above 62% wins with a positive average. Small batches react to recent results, giving less weight to probes that moved far against the trade. An 8 by 8 grid stores outcomes across trend/range and volatility conditions. Its influence grows with supporting observations, up to 65% of the context/global blend.

The model blends proposals into a target and smooths the update. A larger multiplier requires a larger reversal to flip trend; a longer ATR length reacts more slowly to volatility changes. Learning changes signal timing. The account retains its separate position and loss limits.

Adaptive Update Equation

$$\theta_{t+1} = \theta_t + \alpha_t(\tilde{\theta}_t - \theta_t)$$

theta is the current setting; theta with a tilde is its rounded, bounded target; alpha is the update fraction. For a multiplier of 1.40, target 1.50 and alpha = 0.20, the next setting is 1.42. Deadbands ignore small changes; update caps and baseline reversion limit drift. Length remains 5-100 and multiplier 0.5-5.0.

3. Turn the adjusted indicator into a decision

BUY requires a stored fresh low, an upward trend flip, RSI(14) below 30 within 50 candles and at least 10 candles since the last signal. RSI compares smoothed gains and losses on a 0-100 scale. SELL uses a fresh-high reversal, downward flip and recent RSI above 70. Volume-surge filtering and live tick pressure are off; all decisions use confirmed candles.

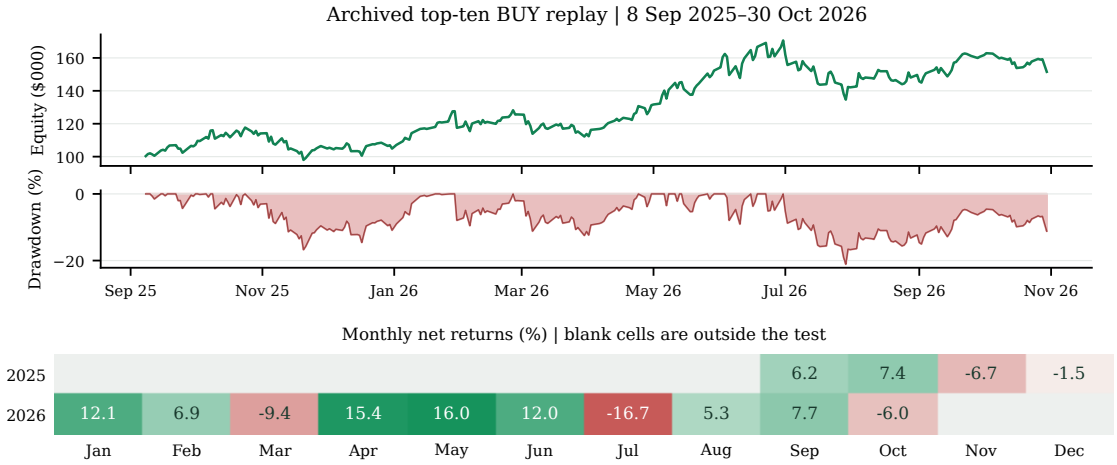
Learning check. On 780 recorded DELL 30-minute bars (2 July-25 September 2026), 776 probes matured per direction. The multiplier ranged from 1.400 to 2.145; ATR length stayed at 24. Adaptive signals were 6 BUY / 8 SELL versus 7 / 10 with fixed settings. Runs ending at candles 200, 400 and 650 reproduced the earlier outputs. Data: Yahoo Finance; our Python calculations. These checks establish model behaviour, not a return advantage.

Dai et al., *Reversals and the Returns to Liquidity Provision* (2023, NBER 30917), link higher volatility to faster, initially stronger reversals. This motivates adaptation; it does not validate our specific entry rule.

2 Archived signals and observed results

Our signal system emailed the ten strongest BUY candidates it found, ranked by score. We archived those dated messages and replayed the saved lists against real historical market prices, rather than recreating picks after seeing subsequent returns. The replay covers 8 September 2025-30 October 2026: 290 sessions, starting with $100,000.

Net total return	Annualized growth	Maximum drawdown	Final marked equity
+51.54%	+43.92%	-21.10%	$151,542.68



Data: archived top-ten BUY emails; Alpaca SIP daily bars, split-adjusted. Our replay and calculations. Costs: 5 basis points of fees plus 5 basis points of adverse slippage per side. September 2025 is a partial month.

The declared strategy exits when a SELL signal appears or a protective stop triggers. The earlier replay uses the first email each date, next-open fills, $10,000 per name and ten holdings maximum. Historical replay exits use ranking turnover, a 5% stop or 15% target. The last email, dated 14 October, is carried forward through 30 October; ten holdings remain marked open.

We reproduced all 1,254 simulated fills: 632 BUYs and 622 SELLs. Dates, tickers, shares, prices, fees and cash amounts match the saved ledger. The results describe the earlier email workflow; the current declaration uses the 30-minute signals and portfolio limits specified here.

Execution sensitivity

Replay case	Total return	Max. drawdown	Sessions
Zero modeled cost	64.45%	-19.47%	290
10 bps per side	51.54%	-21.10%	290
20 bps per side	39.11%	-22.81%	290
10 bps + one extra session	43.49%	-21.91%	289

Doubling modeled costs reduces return by 12.43 percentage points; delaying ranking eligibility reduces it by 8.05 points. The delayed case starts one session later; stop and target exits retain next-open execution. Allocation and exit thresholds are fixed across cases. These results show material sensitivity to execution costs and timing.

The archived-ranking replay returned 51.54% versus QQQ at 19.39%, but used different entry and exit rules. In a separate direct 30-minute test of 17 names (2 Jul-9 Oct 2026; 70 sessions), adaptive settings returned 5.95% (max drawdown -3.71%) versus 12.97% (-3.89%) with fixed ATR(24) / 1.4 settings; a 75% SPY / 25% cash benchmark returned 3.08%. At 10 bps per side, returns were 4.22%, 11.50% and 3.04%. Adaptation did not beat fixed settings in this short, selected sample. Method, source hashes and results: direct replay.

Research repository: github.com/V-Juneja/ml-supertrend
Replay code, dated rankings, result tables, charts and reproduction instructions: docs/BACKTEST.md.

3 Trade expectancy and risk tests

Net Trade Expectancy Equation

$$e = pW - (1-p)D, \qquad p_{BE} = \frac{D}{W+D}$$

e is mean net return per closed trade; p is the net win rate; W is the average net winning return and D the positive magnitude of the average net losing return. All returns use percentage units and already include costs. The break-even win rate, p BE, is the win rate needed to offset losses at those payoff sizes.

Trade statistic	Historical result
Completed trades	622
Win rate	46.62%
Mean net profit / return per trade	+$81.73 / +0.8173%
Median trade return	-0.6268%
Average net winning / losing return	+8.05% / -5.50%
Profit factor	1.28
Break-even win rate at observed payoffs	40.59%
Worst trade	-40.03%
Longest losing streak	15 trades

Winners average 1.46 times losers, so the 46.62% win rate exceeds the 40.59% break-even level. Profit factor 1.28 means $1.28 of positive net P&L per $1 of negative net P&L. The mean gain is $81.73 per $10,000 trade, even though the median trade loses 0.6268%: larger winners lift the average.

The 15-loss streak and -40.03% worst trade show why position limits and gap controls matter. These are 622 closed trades from the earlier email replay; ten open positions are excluded. Streaks follow recorded SELL order, including same-day exits. Calculations: GitHub trade evidence.

Monte Carlo: exposure, correlation and gaps

The separate historical risk sample is distinct from the five-position strategy portfolio. We fit its covariance to 59 aligned daily log returns, 2 July-25 September 2026, with 11.00% Ledoit-Wolf shrinkage. Each model runs 50,000 fifteen-session paths at 40%, 60% and 75% exposure, with fixed quantities and zero cash interest. These scenarios compare risk levels; they are not strategy returns. Expected shortfall (ES) is the average account loss in the worst 5% of paths.

Risk scenario	ES 95% at 40% / 75%	Investment implication
Gaussian / Student-t(5)	3.50% / 6.56%; 3.56% / 6.68%	Zero-drift baseline / heavier tails.
Five-session block bootstrap	2.92% / 5.48%	Preserves sampled return blocks; unseen crashes remain possible.
Volatility x 1.5; correlation 0.90	9.36% / 17.55%	Correlated losses are the dominant portfolio stress.
Common 20% opening gap	10.81% / 20.27%	At 75%, daily-close liquidation lowers ES to 16.44%.

Data: Yahoo Finance daily closes; our calculations. Common dates only, no missing-price fills. Gaussian ES rises from 3.50% at 40% to 6.56% at 75%; the 60% comparison is 5.25%. At 75%, 7.18% of Gaussian paths reach the 6% drawdown trigger. These fixed-holding scenarios test risk, not strategy returns. Gaussian expected price drift is zero; the short calibration leaves regime uncertainty. Seeds: 20261102-20261106.

4 Portfolio construction and risk

False reversals can cluster during persistent downtrends; overnight gaps can cross stops, and technology names can fall together. Entry spacing, planned-loss budgets, sector/theme caps and the account halt address these risks. All limits below apply together and are declared design choices, not optimized thresholds.

Control	Execution rule and purpose
Exposure	Long-only, no borrowing; at most five names, 15% of equity per name and 75% gross stocks. Five names at the individual cap use the full portfolio allowance.
Loss budget	Initial stop two 30-minute ATR(24) below fill; planned risk at most 0.5% of equity per name and 2% across the account. Never widen a stop.
Concentration	At most 20% in one sector and 25% in the combined semiconductor / technology-infrastructure cluster. Correlated names consume shared capacity.
Liquidity	Price at least $5; trailing 20-session average dollar volume at least $20 million. Reject stale, halted or incomplete quotes.
Account losses	At 2% session loss, pause entries until the next session. At 6% from peak marked equity, liquidate at available prices and halt for the competition.
Execution integrity	Use one account; reject duplicate BUY entries. Log accepted/rejected signals, settings, quotes, order and fill times, fees, stop and exit reason. Freeze the declared rules; close by the competition cutoff.

Position Sizing Equation

$$d_i = 2\,\mathrm{ATR}_{24,i}, \qquad q_i = \left\lfloor \min\left(\frac{0.15E}{P_i}, \frac{0.005E}{d_i}\right) \right\rfloor$$

For stock i, E is marked account equity, P its entry price, ATR with subscript 24 its 24-candle smoothed true range, d the dollar stop distance and q whole shares. Floor brackets round down. The smaller value/risk cap governs; remaining exposure, sector capacity, aggregate risk and costs can reduce the order.

At the recorded DELL close of $563.17, a $15,000 cap permits 26 shares ($14,642.42). The $500 loss budget in a $100,000 account allows at most $19.23 stop distance at that size. A wider two-ATR stop reduces shares. The 15% name limit and five-position limit allow at most 75% gross exposure; remaining equity stays in cash. Position size can be lower when the stop-risk budget binds.

Gap and execution stress

Stop execution stress	Loss / planned stop loss	Account loss at 2% risk budget
Fills at intended stop	1.0 x	2%
Loss exceeds stop distance by 50%	1.5 x	3%
Loss doubles the intended distance	2.0 x	4%
Gap produces three times planned loss	3.0 x	6%

Calculation: aggregate planned loss multiplied by the realized loss/stop-distance ratio. This is a deterministic exposure test, not a forecast. Gaps can exceed all ratios shown; neither a stop nor the 6% liquidation trigger guarantees a maximum loss.

Stress grid: 96 combinations of horizon, exposure, volatility and correlation (100,000 paths each); daily Gaussian, t(5), gap and liquidation tests use 40,000 paths per model. At the 75% ceiling, common 10%/20% falls imply 7.5%/15% account losses before costs. A 6% halt is a trigger, not a loss guarantee.

Final declared strategy

Trade confirmed 30-minute ML Supertrend reversals, long-only. Matured probes update bounded settings; qualifying BUYs pass the sizing and portfolio limits above. Hold at most five names, 15% per name and 75% total stocks. Exit on SELL or stop; pause entries at 2% session loss and liquidate and halt at 6% account drawdown.